



# Enhancing quadcopter motor performance prediction using Jaya-optimized feed forward neural network

Chandrasekaran Ravichandran<sup>1</sup> · Raja Balakrishnan<sup>2</sup> · Selvajyothi Kamakshy<sup>3</sup>

Received: 18 September 2024 / Accepted: 2 December 2024 / Published online: 14 January 2025  
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2025

## Abstract

In this paper, a Jaya-optimized feed forward neural network (FFNN) is used to model the brushless DC (BLDC) motors in the quadcopter. Precise modeling of motor-propeller performance parameters for a given throttle input, including speed, thrust, current, torque, and power is necessary for estimating energy consumption and optimizing energy sources for various flying conditions. Accurate modeling is essential for performance analysis, but it is challenging due to the nonlinear relationships, environmental factors, and variability in propeller and motor characteristics. However, by integrating experimental data with simulation and analytical models, the prediction accuracy can be enhanced. While FFNNs are effective in capturing the complex nonlinear relationships between input and output variables, their performance is highly dependent on the optimization of network weights and biases. Traditional optimization methods often face challenges such as overfitting, computational inefficiency, and sensitivity to initial conditions. By leveraging the Jaya algorithm, which requires minimal control parameters and promotes adaptive convergence, the FFNN's predictive accuracy is significantly improved. In this research, the performance of the model is studied and verified experimentally with two different throttle conditions, namely 1). linearly increasing and 2) full flight from takeoff to landing. The following performance parameters speed, thrust, current, torque, and power are predicted with an accuracy of 96.4%, 97.5%, 97.2%, 94.9%, and 96.8%, respectively, at maximum throttle positions with respect to traditional models.

**Keywords** Jaya algorithm · Optimization · Feed forward neural network · Regression model · Performance prediction · Quadcopter motor performance modeling

## 1 Introduction

The quadcopter has wide applications and a few to mention are military surveillance, law enforcement, geographic mapping, wildlife monitoring, precision agriculture, rescue operations during national disasters and other calamities, delivery of packages, photographing and analysis, and transmission line monitoring [1]. Recently, it has been found in applications in entertainment, shipping, and delivery of goods [2]. The wide application is attributed to the versatility of quadcopters in flying across different conditions and areas [3]. Unlike fossil fuel drones, electrified quadcopters with lightweight material construction have superior maneuverability and zero emission potential. They can also be equipped with essential additional components such as cameras, global positioning system (GPS)-guided rockets, navigation systems, and sensors. Based on the distance covered and endurance, the drones are classified as short-range

✉ Chandrasekaran Ravichandran  
edm20d003@iiitdm.ac.in

Raja Balakrishnan  
rajab@iiitdm.ac.in

Selvajyothi Kamakshy  
ksjyothi@iiitdm.ac.in

<sup>1</sup> Department of Electronics and Communication Engineering, Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, Chennai, Tamil Nadu 600127, India

<sup>2</sup> Department of Mechanical Engineering, Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, Chennai, Tamil Nadu 600127, India

<sup>3</sup> Department of Electronics and Communication Engineering, Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, Chennai, Tamil Nadu 600127, India

(< 10 km, 1 h), medium-range (< 500 km, < 24 h), and long-range (> 500 km, > 24 h) [4].

The mathematical modeling of quadcopter is an important aspect of the design and development of closed-loop control. The robustness of the developed controller is improved by considering the effect of external disturbance, delay in communication and signal processing, and the life cycle of the quadcopter [5]. There are various mathematical models used in practice. The rigid body dynamics of the quadcopter have been modeled by using the Euler–Newton approach or Lagrangian approach [6]. Accurate forward flight conditions have been modeled and analyzed with hub forces, rolling moments, and variable aerodynamical coefficients [7]. The stable altitude control using back-stepping integral control has been implemented using blade element and momentum theory [8]. The effect of wind disturbances on attitude and altitude control in quadcopters at high velocity is modeled and validated in an indoor thrust stand [9]. System identification models are created for the quadcopter using test data, including the development of an autoregressive exogenous (ARX) input model based on altitude data and a multiple input multiple output (MIMO) model derived from angular velocity data. These models are specifically designed for the Mark III quadcopter system, utilizing experimental flight data [5]. To improve the accuracy of the predictions, the Box–Jenkins (BJ) identification model has been used [10].

A second-order frequency response method has been used to model the roll, pitch, and yaw control [11]. The effectiveness of individual motors has been modeled using the Fourier transform regression (FTR) method [12]. An online aerodynamic parameter identification has been modeled using a recursive least squares algorithm [13]. There are combined rotor aerodynamics and experimental data based Gray–Box models [14]. The DeePC algorithm is used to control the quadcopter [15]. The artificial neural network (ANN)-based regression model is used to achieve high maneuverability, robust control, and stability in quadcopter with environmental changes and plant parameter variations [16]. Modeling the uncertainties in quadcopter has been a complex task, and the ANNs can be applied for this problem. The application of ANN in quadcopter is classified into two groups: 1) ANN-based compensator/estimator and 2) neural network-based regression modeling.

The system identification and control of the quadcopter is modeled using general regression neural network (GRNN) [17]. A black-box model using an radial basis function neural network (RBF-NN) is formulated with the minimum resource allocation network (MRAN) approach to control the longitudinal dynamics of a quadcopter [18]. The gated recurrent units (GRUs), convolutional neural networks (CNNs), and long short term memory (LSTM) have been employed to develop single input single output (SISO), and MIMO models were implemented and experimentally verified in [19].

The servo motor model and roll rate model are identified from frequency response method [20] to control the dynamics and trajectory tracking. The particle swarm optimization (PSO) algorithm is used to optimize the structure and number of inputs to predict the daily gas consumption in [21] sound navigation and ranging (SONAR) target classification [22]. The Jaya algorithm is the optimization methods of ANN in [23] suggests objective functions such as (i) initialization of weights, (ii) number of iterations, (iii) volume of the training set, (iv) learning rate, and (v) number of hidden layers. Different methods have been applied to the architectural configuration and learning or training of ANN.

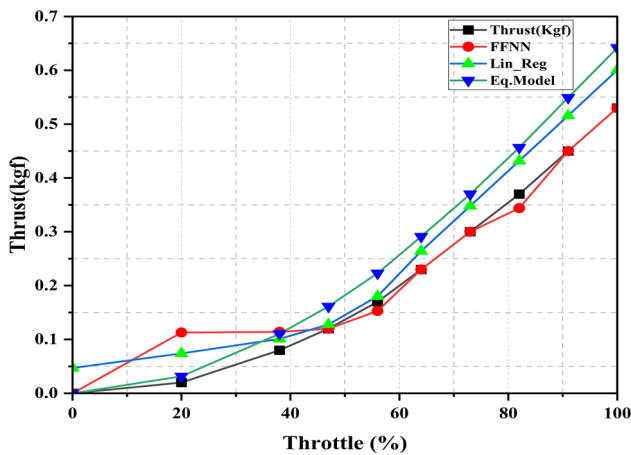
A virtual test bench is an important requirement in the development and testing of electric transportation systems, including electric vehicles (EVs) and the supporting infrastructure [24]. The development of a virtual test bench involves collaboration among multidisciplinary teams of engineers, simulation experts, and domain specialists [25]. It is an important tool for accelerating the development and testing of electric transportation systems and also offers numerous advantages, including cost savings, safety, time, efficiency, and the ability to simulate various scenarios [26]. Artificial Intelligence (AI) techniques play a significant role in enhancing the capabilities and effectiveness of virtual test benches for electric transportation systems [27]. Gravitational search algorithm is applied to train the FFNN and compared with other algorithms in [28]. Incorporating AI into quadcopter motor modeling not only enhances their capabilities but also accelerates the development and testing of electric transportation systems, reduces costs, and improves the overall reliability and safety. It allows engineers to simulate and evaluate complex scenarios more effectively, leading to efficient and innovative solutions in the electric mobility industry.

The main contributions of this research work are:

- Modeling the quad copter motor using FFNN.
- Optimization of weights and biases of FFNN using the Jaya algorithm to minimize the error between actual and estimated motor speed, thrust, current, torque, and power.
- Validation of the Jaya FFNN model with experimental data from a BLDC motor and for a real-time flight scenario.
- Comparison of the proposed Jaya FFNN model with other existing models such as PSO-FFNN, linear regression, neural network regression (NN-Reg), and SVM.

## 1.1 Problem statement

Brushless DC (BLDC) motors are widely used in various applications due to their high efficiency and reliable performance. Modeling the nonlinear behavior of the BLDC motor in a quadcopter application is crucial to evaluating the speed,



**Fig. 1** Comparison of thrust (kgf) versus % throttle position using experiment, theoretical, FFNN model, and linear regression model

thrust, torque, and power of the given throttle input. Also to include the outdoor effects on an indoor thrust stand requires additional equipment and increases the installation cost. The ANN models are well suitable for formulating the nonlinearities in plant input–output characteristics. Training an ANN model to predict the motor performance in a quadcopter is a complex task and requires a large set of test data. Conducting the thrust stand experiments for different throttle input conditions to collect the datasets of electrical and mechanical parameters such as voltage, current, throttle, thrust, torque, input power, and output power is time-consuming and expensive process. The feed forward neural networks (FFNNs) have shown promise in capturing the complex nonlinear relationships between the throttle input and motor performance with minimum test dataset and improved accuracy. However, the effectiveness of FFNNs solemnly depends on the selection of their weights and biases. Traditional optimization methods struggle with the high-dimensional non-convex solutions. The higher degree of nonlinearities in this problem leads to the suboptimal solutions. Therefore, it requires a robust optimization approach that can effectively train FNNs to predict the performance of BLDC motor accurately. Figure 1 shows the nonlinear behavior of the motor's thrust response to the throttle input. The prediction of FFNN regression model over-fits the data due to the random choice of

FFNN architecture and the weights. Further, there are no standard methods to choose the number of layers, the number of nodes in each layer, the activation function, and weights used in each node of the ANN model. Hence, determining the optimal weights of the ANN network is challenging and often requires trial and error methods. To overcome the above-mentioned complexities and to enhance the prediction accuracy of the model, a Jaya algorithm is used to optimize the weights of the FFNN model.

## 1.2 Simple BLDC motor model

By directly influencing the voltage and current supplied to the motor, throttle position determines the speed, torque, and power consumption, impacting overall quadcopter dynamics. Understanding these relationships through basic electrical and mechanical principles provides a solid foundation for optimizing motor performance in various flight scenarios. To begin with let's consider the modeling of a BLDC motor. The electronic speed controller (ESC)-motor subsystem model shown in Fig. 2 is used to regulate the motor's angular velocity  $\omega$  to achieve the desired angular velocity  $\omega_d$ . The model is composed of the following components: a series inductor of inductance  $L_a$  and a resistor of resistance  $R_a$ . This model helps to understand the electromechanical dynamics of the rotating machinery. The input voltage of the brushless DC (BLDC) motor is regulated by PWM control command given to ESC. For the given command, the motor responds with angular velocity  $\omega$  and motor current  $I$ .

The average voltage applied to the motor through the throttle and ESC is given by Eq. (1)

$$V_{avg} = V_b * D \quad (1)$$

where,  $D$ —% Duty cycle of the PWM control pulse.  $V_{avg}$ —effective voltage from the ESC.  $V_b$ —battery voltage.

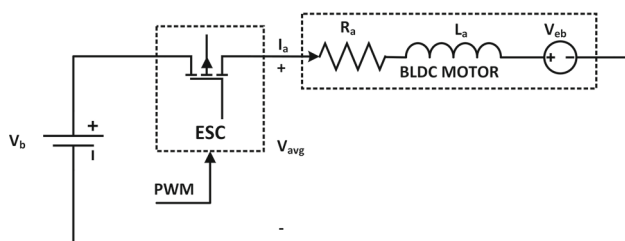
The throttle position also influences the current supplied to the motor, which directly affects the torque produced. This relationship is crucial for the motor's performance, especially in terms of acceleration and load handling. The back EMF generated by the motor's motion opposes the applied voltage. The current drawn by the motor is given as

$$I_a = \frac{V_{avg} - V_{eb}}{R_a} \quad (2)$$

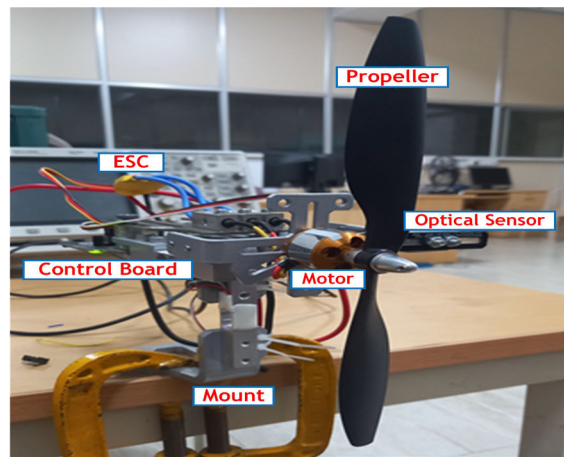
$I_a$ —Current through the motor.  $V_{eb}$ —Back electromotive force generated by the motor.  $R_a$ —Resistance of the motor windings.

The speed of a motor is dependent on the applied voltage when the magnetic flux is constant. Therefore,

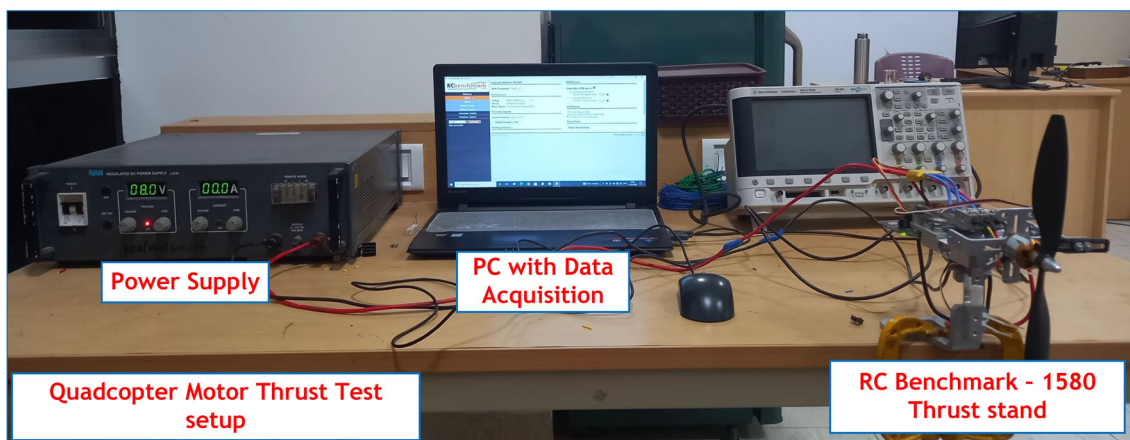
$$V_{eb} = K_e \omega \quad (3)$$



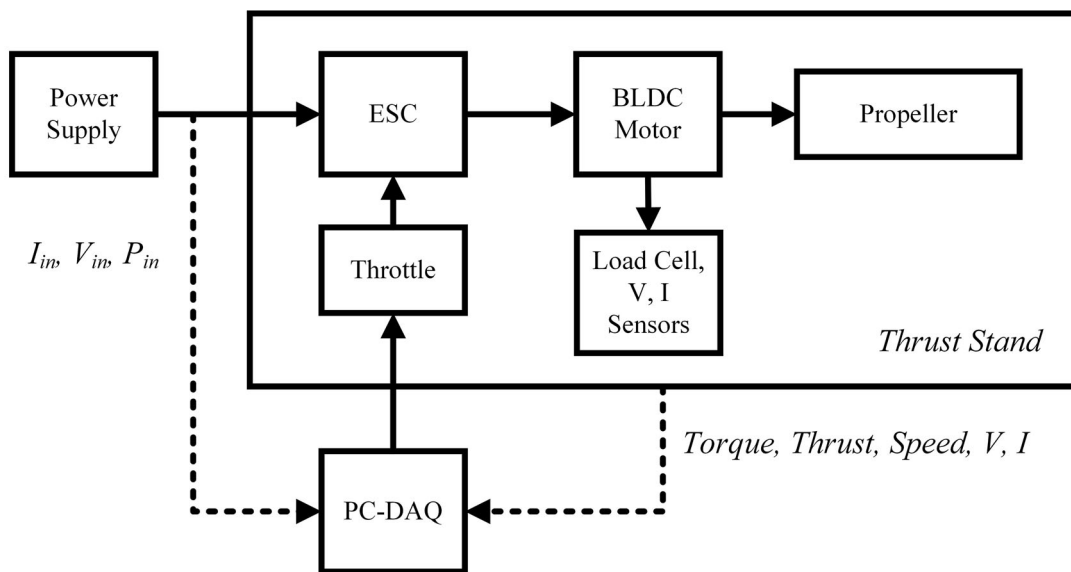
**Fig. 2** ESC-motor electrical equivalent circuit model



(a)



(b)



(c)

**Fig. 3** a. Configuration of thrust stand with motor, propeller, ESC, and control board. b. Test bench demonstrating the data acquisition from the thrust stand. c. Block diagram representation of the data acquisition from the thrust stand

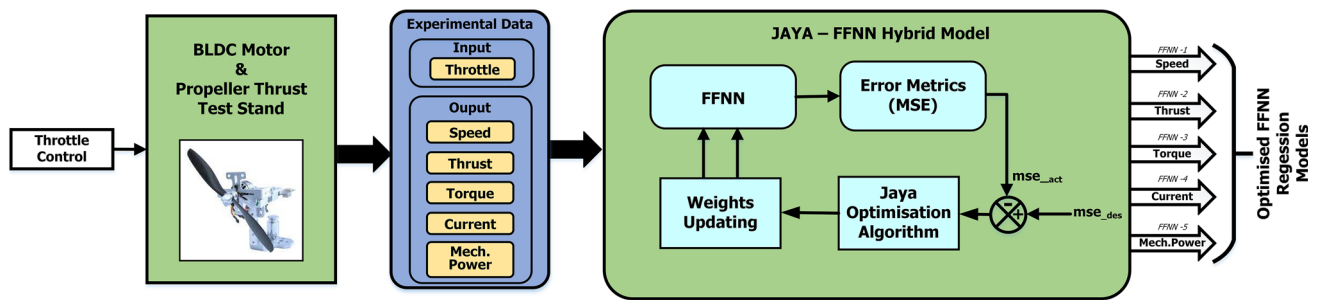


Fig. 5 Overall schematic of the proposed Jaya-optimized FFNN model

Fig. 6 Architecture of the proposed FFNN

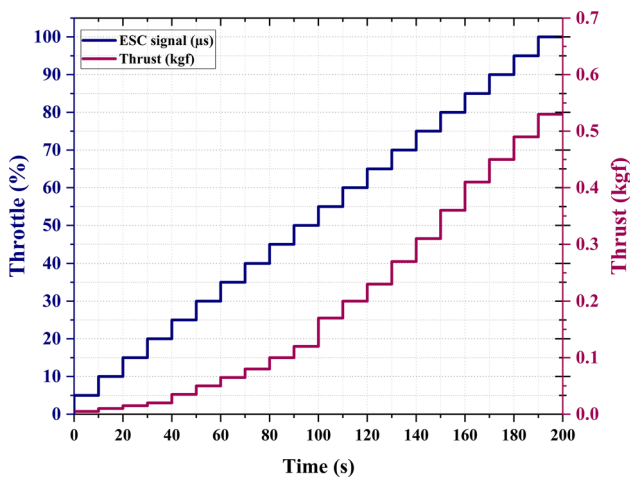
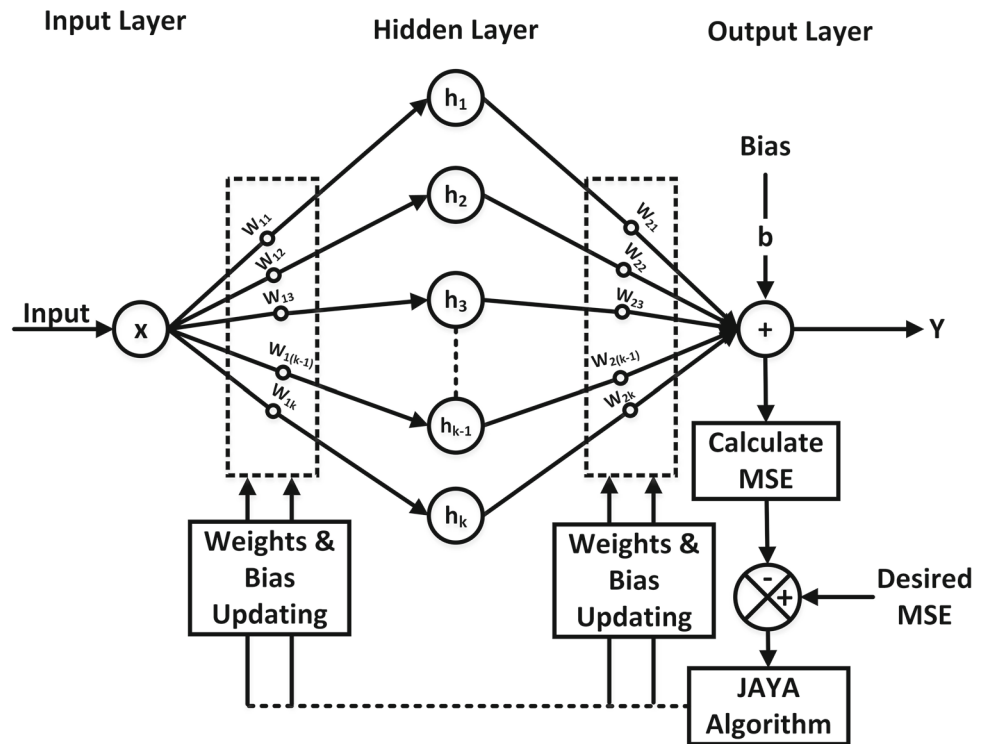


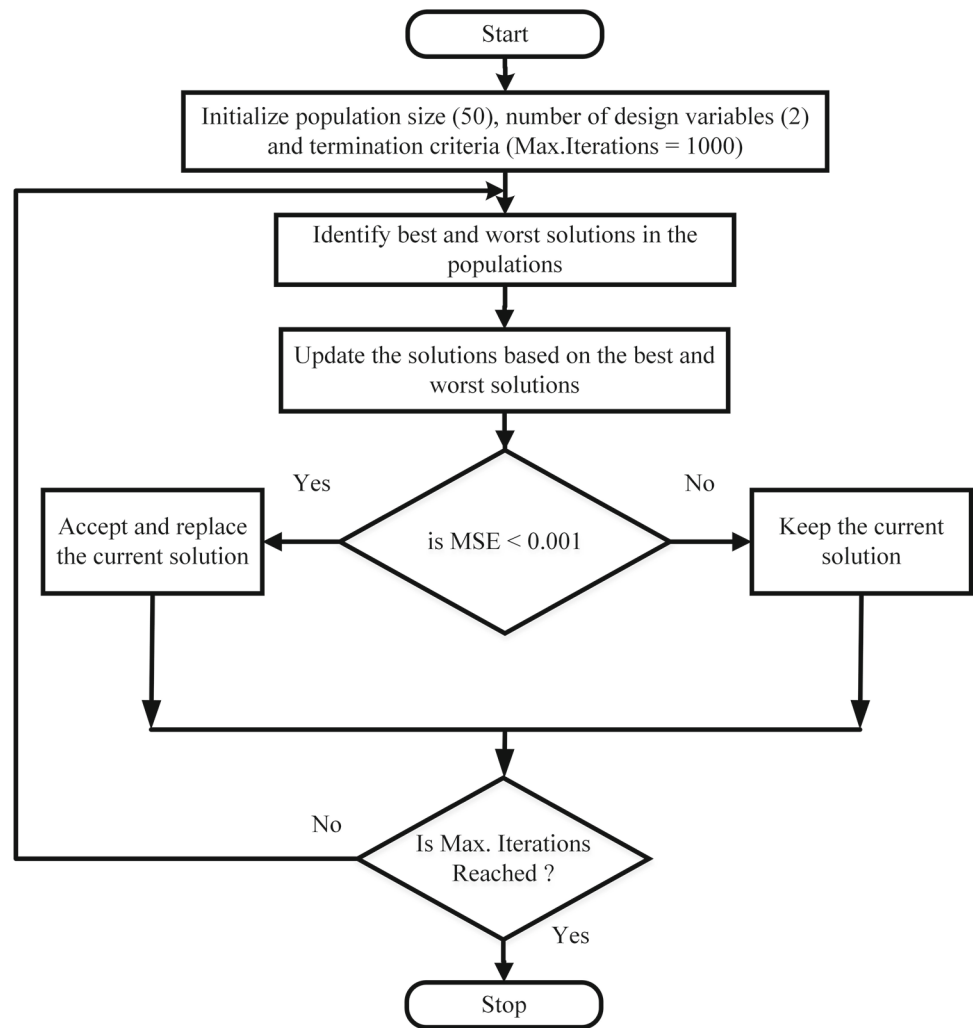
Fig. 4 Step variation of % throttle and thrust with respect to time

Table 1 Specification of the BLDC motor used in this study

| S.No. | Parameter                         | Values      |
|-------|-----------------------------------|-------------|
| 1     | Current at maximum efficiency     | 6–12 A      |
| 2     | No load current                   | 0.5A @ 10 V |
| 3     | Peak current                      | 16 A/60 s   |
| 4     | Maximum efficiency                | 80%         |
| 5     | Internal resistance ( $R_{int}$ ) | 90mΩ        |
| 6     | Number of poles                   | 14/12       |
| 7     | Maximum power                     | 180 W       |
| 8     | Propeller                         | 10" × 4.5"  |
| 9     | Motor speed constant              | 1000 kV     |

where,  $K_e$  is the counter EMF constant measured in V-s/rad,  $\omega$  - angular speed of the motor (rad/s).



**Fig. 7** Flow chart for Jaya algorithm**Table 2** Specification of the BLDC motor thrust stand

| S.No. | Parameter        | Values |
|-------|------------------|--------|
| 1     | Thrust           | 5 kg-f |
| 2     | Torque           | 2 Nm   |
| 3     | Voltage          | 50 V   |
| 4     | Current          | 55 A   |
| 5     | Measurement rate | 80 S/s |

The fundamental relationship between voltage and motor speed for a BLDC motor can be expressed by:

$$\omega = \frac{V_{avg} - I_a R_a}{K_e} \quad (4)$$

The torque ( $T$ ) produced by the BLDC motor is given by:

$$T = K_t I_a \quad (5)$$

where,  $T$  is the motor torque (Nm),  $K_t$  - torque constant (Nm/A), often equal to the reciprocal of the velocity constant.

Higher throttle position increases the duty cycle of the PWM (Pulse width modulation) signal from the ESC, allowing more current  $I$  to flow through the motor windings, resulting in higher torque  $T$ . The electrical power  $P$  consumed by the motor is given by:

$$P = V_{avg} I_a \quad (6)$$

and in terms of mechanical power is given in Eq. (6)

$$P_{mech} = T \omega \quad (7)$$

The thrust  $F$  generated by the propeller is related to the square of the rotor speed and is given by:

$$F = C_T \rho A \omega^2 \quad (8)$$

where,  $C_T$  is the thrust coefficient,  $\rho$  is the air density ( $\text{kg/m}^3$ ),  $A$  is the propeller disk area ( $\text{m}^2$ ),  $\omega$  is the angular speed ( $\text{rad/s}$ ), and Eq. (8) can be rewritten as

$$F = k_{th}\omega^2 \quad (9)$$

where,

$$k_{th} = C_T * \rho * A$$

Higher throttle increases both voltage  $V$  and current  $I$  drawn from the battery source leads to higher power consumption, derating the thermal performance and impacting battery life. By balancing throttle position for optimal performance without excessive power draw, it will ensure efficient battery performance. The thrust produced by the propeller is highly nonlinear with respect to speed and the throttle position. Hence, by understanding the relationship between throttle position and motor performance, Jaya-optimized FFNN-based data-driven model can accurately predict and optimize the performance of a BLDC motor. This approach is particularly valuable in applications like quadcopters, where precise control and efficiency are essential.

## 2 Methodology

This study aims to develop a highly accurate model for predicting quadcopter motor performance using a Jaya-optimized feed forward neural network (FFNN). To achieve this, an experimental setup was established for data acquisition, followed by the design and optimization of an FFNN model using Jaya optimization. The methodology includes data collection through a static thrust test, the development of the optimized neural network model, and comparisons with alternative modeling approaches to validate the accuracy and robustness of the proposed method.

### 2.1 Experimental setup

A static thrust test for the motor-propeller pair was conducted using the RC Benchmark—1580 thrust stand, as shown in Fig. 3a. The static thrust test does not include the environmental factors like wind, temperature, and humidity, as they do not immediately affect motor performance under controlled conditions. However, environmental changes, like wind resistance and temperature fluctuations, could indirectly impact energy consumption, battery performance, and motor efficiency in real-world scenarios. All experiments were conducted in laboratory under the standard temperature of 25 °C. To ensure precise measurement, the thrust stand was calibrated before each experiment. The data acquisition setup using the thrust stand is shown in Fig. 3b. The block

diagram representation of the thrust stand along with the data acquisition is shown in Fig. 3c. Table 1 shows the specifications of the BLDC motor used for testing, and Table 2 shows the specifications of the RC Benchmark—1580 quadcopter thrust stand BLDC motor.

The test bench is powered by 12 V source drawing a current of 10 A to excite the BLDC motor through the electronic speed controller (ESC). The throttle is initially set to 5% and increased at an interval of 5% till 100% is reached. The data are acquired at a rate of 10 S/s. At each step, there is a defined settling time for the system to reach a steady state as shown in Fig. 4. Once the system settles with constant thrust values then, all the data incorporating thrust, speed, voltage, current, torque etc. are captured and stored. This process repeats until the maximum throttle is reached.

### 2.2 Proposed Jaya-optimized FFNN model

Figure 5 illustrates the proposed Jaya-optimized FFNN model for the BLDC motor. The data acquired from the thrust stand for various throttle positions are fed into the model. While FFNNs are effective in capturing the complex nonlinear relationships between input and output variables, their performance is highly dependent on the optimization of network weights and biases. Traditional optimization methods often face challenges such as overfitting, computational inefficiency, and sensitivity to initial conditions. The Jaya optimization algorithm, with its ability to efficiently navigate complex error surfaces and its algorithm-specific parameters, offers a robust solution to these challenges.

The model was trained using 4236 data points that include various operating conditions, from minimum to maximum control inputs, as well as dynamic scenarios like sudden ascents and descents, which provided a comprehensive dataset for accurately learning the relationships between input throttle commands and output parameters like speed, thrust, torque, and power. These data points were gathered under various conditions to capture a wide range of motor performance metrics. This enables the model to better predict and adapt to real-time changes in flight dynamics. Eighty percent of the collected data is used for training the model and 20% used for validation. The developed FFNN model is used to demonstrate the flight performance, including speed, thrust, torque, current, and mechanical power propulsion for different throttle inputs, and the model was tested to assess the effectiveness and the prediction accuracy.

To ensure the accuracy in prediction during dynamic changes in data patterns due to the frequent and unpredictable shifts, the adaptive Jaya FFNN model with online learning method is proposed. The key feature is a feedback loop that monitors model performance, automatically triggering adjustments if accuracy begins to decline. This feedback

mechanism includes a thresholding system, where different levels of accuracy loss prompt targeted responses minor weight adjustments for slight deviations and full retraining for larger shifts. This keeps the model focused on the most recent data while minimizing reliance on outdated trends. To ensure the long-term adaptability, periodical full retraining on a larger dataset is suggested to capture any substantial shifts in prediction due to aging effect.

### 2.3 Architecture of Jaya-optimized FFNN

Feed forward neural network (FFNN) is used to develop quadcopter motor-propeller model to predict the performance under different flight conditions. The proposed Jaya FFNN model consists of an input layer, a hidden layer, and an output layer as shown in Fig. 6. The FFNN models with less number of hidden layers are a good choice for regression models, and they reduce the complexity in the optimization process. Furthermore, neurons in every layer are connected using synaptic links known as weights ( $w$ ) of different neurons, together with bias ( $b$ ) parameters in the previous layers. Mathematically, the output of this architecture is expressed as in Eq. (10)

$$y = \sum_{i=1}^k \omega_i x_i + b \quad (10)$$

The Lavenberg Marquardt (LM) algorithm is used to train the FFNN. The LM algorithm is a complex gradient algorithm used to improve the effectiveness of the FFNN. The weight and bias of the nodes are adjusted until the mean squared error (MSE) of the outputs is minimized to a threshold level. This is achieved with the help of an optimization algorithm formulated as in Eqs. (11) and (12).

Minimize

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2 \quad (11)$$

$$\text{Subject to, } -1 \leq \omega_{ij} \leq 1 \text{ and } 0 < b \leq 1 \quad (12)$$

When a FFNN is designed, two major problems are encountered: (a) determining the best topology of its architecture, i.e., the total hidden layers and total nodes in it, and (b) optimizing the initial weights of the training nodes. There is no standard procedure for assigning the number of neurons in the hidden layer; in general, it is done using the trial and error method [29]. The FFNN model contains 5–20 neurons with single hidden layer as in [30] assures the high accuracy in predicted results. Therefore, in this proposed model, the number of neurons in the hidden layer is chosen as 5, 10, 15, and 20 to find the better prediction model. The most

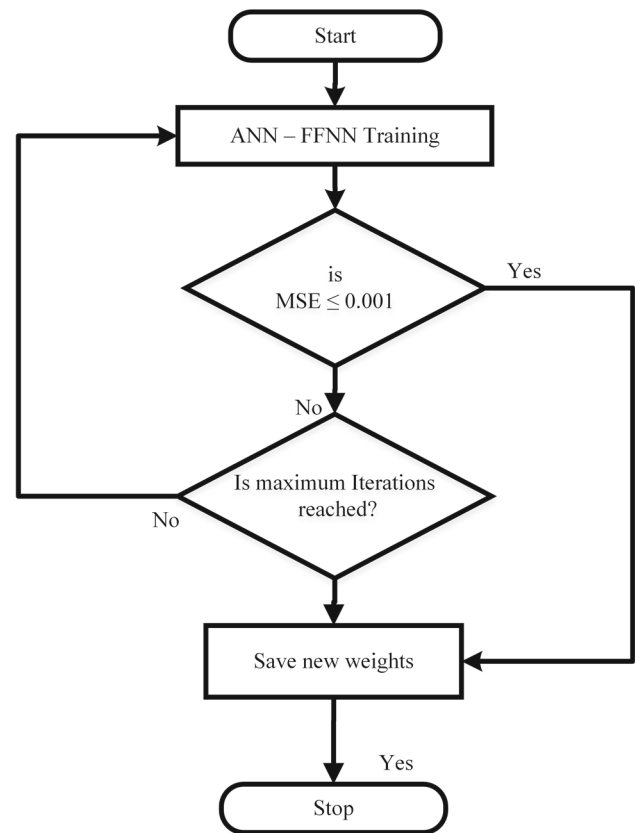


Fig. 8 The proposed training scheme for ANN

common FFNN activation functions are tansig (tangent sigmoid), logsig (logistic sigmoid), and linear functions. From the initial trials and previous studies, the tansig and linear activation functions are opted for the input/hidden layers and the hidden/output layers, respectively.

### 2.4 Jaya optimization algorithm

The Jaya algorithm is a population-based optimization technique, and the control parameters are the population size and generation count [31]. The Jaya algorithm has a simple numerical structure and only one phase of execution which makes the coding and execution very simple. Unlike other metaheuristic algorithms, the Jaya algorithm is not prone to becoming trapped in local minima [32] because in this algorithm, the solution evolves from the global best values.

As shown in Fig. 7 of the Jaya flowchart, the algorithm generates an initial population at random and determines the best and worst solutions based on the objective function of the given problem. If  $X_{j,k,i}$  is the value of the  $j$ th variable for the  $k$ th candidate during the  $i$ th iteration, then the MSE of the Jaya-optimized FFNN model is obtained as in Eq. (13)

$$X'_{j,k,i} = X_{j,k,i} + r_{1,j,i} (X_{j,\text{best},i} - |X_{j,k,i}|)$$



**Table 3** Convergence analysis of test runs for Jaya-optimized FFNN model

| Test No. | Execution time (s) | MSE        | Time @ optimum MSE (s) | Iteration @ optimum MSE |
|----------|--------------------|------------|------------------------|-------------------------|
| 1.       | 144.6265           | 0.00100155 | 36.73513               | 254                     |
| 2.       | 148.9193           | 0.00114663 | 47.8031                | 673                     |
| 3.       | 148.7318           | 0.00100205 | 10.55996               | 71                      |
| 4.       | 142.0111           | 0.00112704 | 39.33707               | 277                     |
| 5.       | 140.6384           | 0.00065759 | 19.83001               | 141                     |
| 6.       | 141.253            | 0.00101752 | 34.18323               | 242                     |
| 7.       | 141.4233           | 0.00101526 | 52.89231               | 374                     |
| 8.       | 141.2432           | 0.00103642 | 58.61593               | 950                     |
| 9.       | 141.5579           | 0.00107551 | 13.87267               | 98                      |
| 10.      | 141.7054           | 0.00102996 | 48.32154               | 341                     |
| 11.      | 142.1262           | 0.00120989 | 11.65435               | 82                      |
| 12.      | 141.894            | 0.00115914 | 36.46676               | 257                     |
| 13.      | 142.057            | 0.00103322 | 27.98523               | 197                     |
| 14.      | 141.8075           | 0.00106014 | 68.35122               | 482                     |
| 15.      | 143.2962           | 0.00042161 | 21.35113               | 149                     |
| 16.      | 144.315            | 0.00106600 | 9.09185                | 63                      |
| 17.      | 143.3652           | 0.00100599 | 13.76306               | 96                      |
| 18.      | 142.6882           | 0.00107428 | 50.08356               | 351                     |
| 19.      | 143.5327           | 0.00100298 | 11.48262               | 80                      |
| 20.      | 161.3096           | 0.00102842 | 28.87442               | 179                     |
| 21.      | 167.3358           | 0.00110478 | 115.62904              | 691                     |
| 22.      | 145.7847           | 0.00101083 | 11.07964               | 76                      |
| 23.      | 312.9977           | 0.00098487 | 165.57578              | 529                     |
| 24.      | 143.4758           | 0.00101799 | 43.04274               | 300                     |
| 25.      | 316.5782           | 0.00103905 | 16.14549               | 51                      |
| 26.      | 191.3058           | 0.00101000 | 18.17405               | 95                      |
| 27.      | 146.3465           | 0.00105141 | 22.24467               | 152                     |
| 28.      | 155.8812           | 0.00103439 | 70.61418               | 453                     |
| 29.      | 256.3305           | 0.00103580 | 165.33317              | 786                     |
| 30.      | 150.5207           | 0.00102210 | 37.93122               | 252                     |

$$-r_{2,j,i}(X_{j,\text{worst},i} - |X_{j,k,i}|) \quad (13) \quad \text{continues until the error has minimized to the threshold value (0.001) or the maximum iteration count (1000).}$$

where  $X_{j,k,i}$ —Optimized solution from the previous iteration,  $X'_{j,k,i}$ —optimized solution of the current iteration,  $X_{j,\text{best},i}$ —global best solution,  $X_{j,\text{worst},i}$ —global worst solution,  $i$ —iteration counter,  $j$ —design variable,  $k$ —population counter,  $r_1, r_2$ —random variables  $0 < r_1, r_2 \leq 1$ .

The algorithm continuously seeks out suboptimal solutions (failure) and moves toward ideal ones (success), resulting in modifications and updates. If the MSE of the new solution is better than the previous best solution, then the current best solution updates the previous best solution; otherwise, the previous best solution is retained. This process

## 2.5 Performance indices of the proposed model

The choice of training algorithm depends on the specific problem, the size of the dataset, and the characteristics of the neural network architecture [33]. Different algorithms are applied to train the neural network and to find the best one. In this research, FFNNs have been trained with Levenberg–Marquardt algorithm, the optimization of weights and bias in the FFNN is optimized using PSO also and compared to the proposed method. To assess the accuracy of the trained FFNN model, the regression metrics such as root mean square

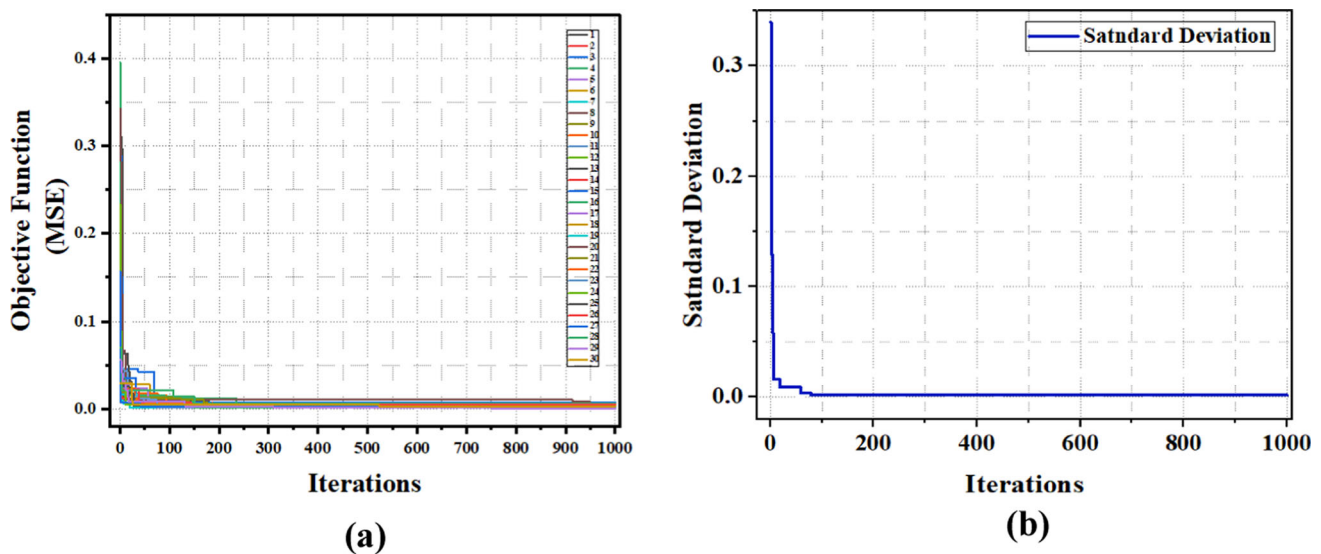


Fig. 9 a. Comparison of MSE convergence for 30 test runs. b. Convergence of standard deviation

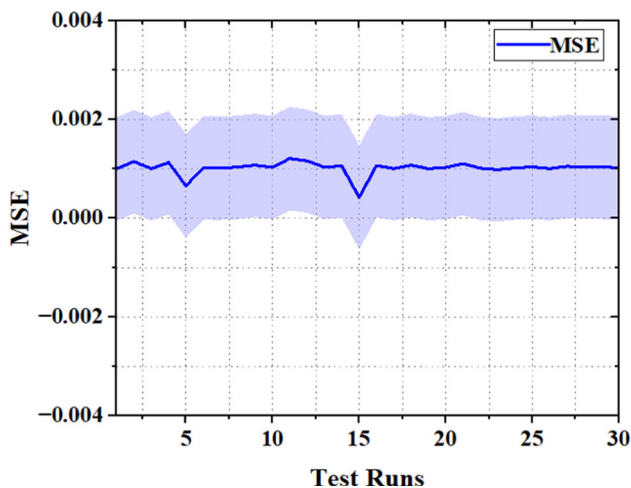


Fig. 10 MSE with standard deviation

errors (RMSEs), mean absolute errors (MAEs), and correlation coefficients ( $R^2$  values) are calculated for the predicted results as in Eqs. (14), (15), (16) [34].

$$\text{RMSE} = \left[ \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2 \right]^{1/2} \quad (14)$$

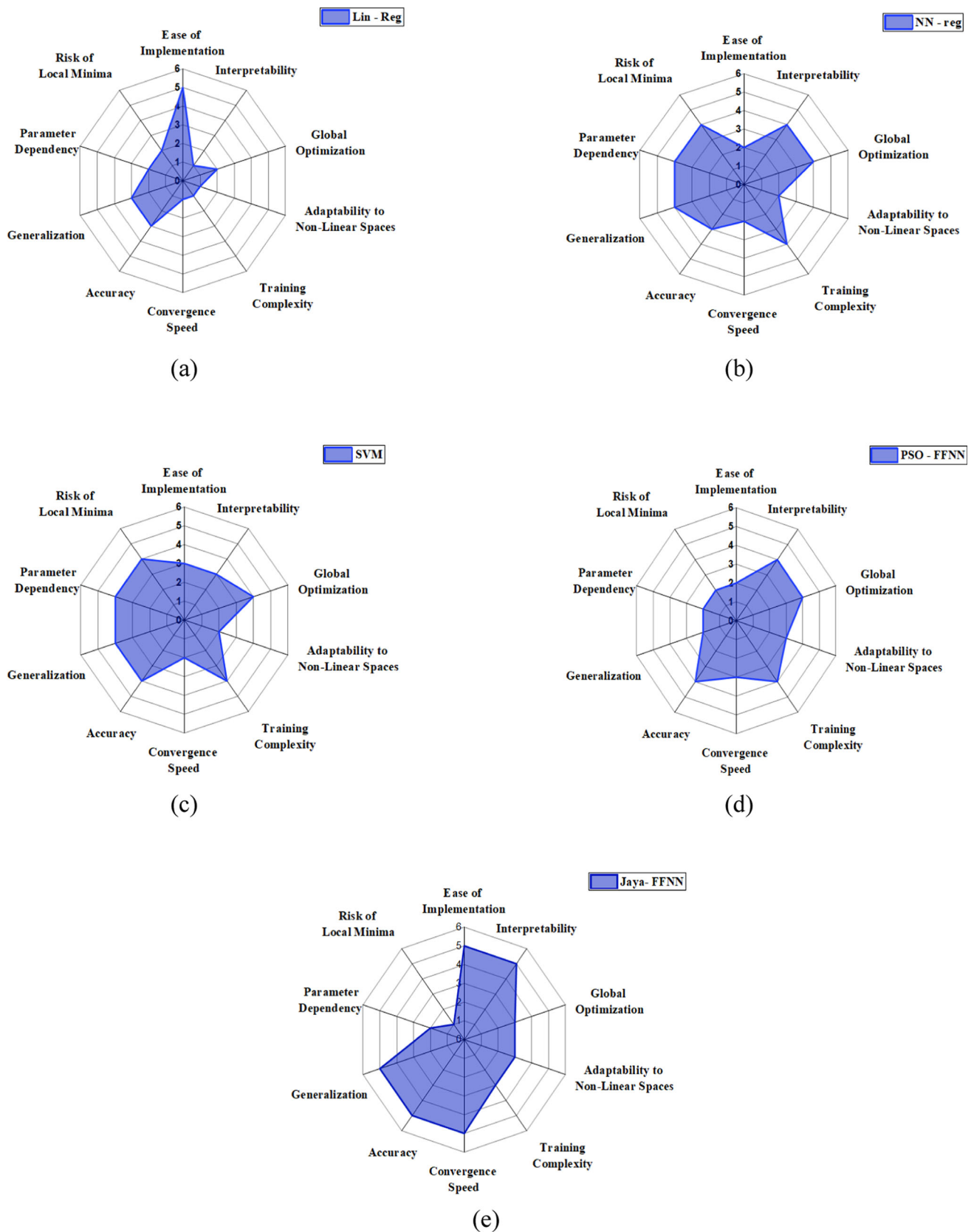
$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - x_i| \quad (15)$$

$$R^2 = \frac{(\sum_{i=1}^n \bar{y}_i - \bar{x}_i)(y_i - x_i))}{\sqrt{\sum_{i=1}^n \bar{y}_i - \bar{x}_i)^2 + \sum_{i=1}^n (y_i - x_i)^2}} \quad (16)$$

In Eqs. (14), (15), (16),  $y_i$  &  $\bar{y}_i$  represents the predicted and mean values and  $x_i$  &  $\bar{x}_i$  represents the real-time input and its

Table 4 Comparison between Jaya and PSO algorithm

| Criteria               | Jaya                                                                                            | PSO                                                   |
|------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Convergence rate       | Faster and consistent                                                                           | Fast initially, may slow in later stages              |
| Stability              | Highly stable                                                                                   | Depends on parameter tuning                           |
| Divergence             | Less prone, can be slow to escape                                                               | Strong in early stages, may get stuck in local optima |
| Reliability            | Consistent and reliable                                                                         | Parameter-dependent, can vary widely                  |
| Ease of implementation | Simple, no parameter tuning is required                                                         | Simple, but requires to tune few parameters           |
| Training complexity    | Lower complexity, faster for many problems                                                      | Higher complexity                                     |
| Parameter dependency   | Parameter free, simpler setup                                                                   | More parameters, higher tuning overhead               |
| Risk of local minima   | Avoids local minima effectively. Converge quickly to suboptimal solutions in complex landscapes | Possible to stuck with local optima if not tuned well |



**Fig. 11** **a.** Performance linear regression model. **b.** Performance neural network regression model. **c.** Performance of support vector machine (SVM) model. **d.** Performance of PSO–FFNN model. **e.** Performance of Jaya FFNN model

mean values, respectively. Smaller RMSE and MAE indicate that a model is generating more accurate estimates and the predictions are closer to the experimental results. Also, the closer the  $R^2 \approx 1$  represents the higher accuracy of the model.

The control parameters of optimization algorithms have major influence on their efficiency, and choosing the right parameters is a difficult task. Based on the experiments, the maximum iteration is set to be 1000 and the number of population is chosen as 100. The target of mean squared error (MSE) was set at  $10^{-3}$ , while the parameter range for the unknown FFNN weights was set to  $[-1, 1]$ . A training procedure as shown in Fig. 8 is used to apply a set of input vectors to guarantee the network's flexibility. This training process continues until the stop criteria are reached. Table 3 shows the convergence speed and stability of the Jaya algorithm for 30 test runs as follow.

The prediction accuracy of the optimized FFNN model has been evaluated from the MSE values of each test run. The convergence speed and stability of the optimization process are used to assess its performance. The convergence analysis is demonstrated as shown in Fig. 9.

The stability of Jaya optimization algorithm is analyzed in terms of the MSE values of each test run, and the standard deviation among the optimum values of the each test run is plotted in Fig. 10. It is observed that the average deviation of MSE value is 0.001.

## 2.6 Limitations and advancements with Jaya-optimized FFNN

Linear regression (LR) model struggles with high errors in predicting thrust and torque for quadcopter motors due to its inability to capture complex, nonlinear relationships [35]. Neural networks (NNs) offered a way to model these nonlinear patterns [36]. Still, they required careful tuning to balance overfitting and under fitting, with limited data and simpler architectures leading to instability under high torque and thrust demands. Support vector machines (SVMs) faced challenges with high-dimensional quadcopter data, and its kernels were insufficient for representing rapid parameter changes, reducing prediction accuracy [37]. Particle swarm optimized feed forward neural networks (PSO-FFNNs) brought robust optimization but were sensitive to tuning parameters and sometimes converged on local minima, failing to fully capture dynamic motor variability, particularly during rapid thrust and torque shifts [38]. Figures 11a–e visually compare these performance attributes, emphasizing Jaya FFNN's efficiency and practical utility in complex real-world environments.

Unlike PSO, the Jaya optimization algorithm improves the feed forward neural network (FFNN) by seeking globally optimal solutions without extensive hyper parameter tuning, which enhances efficiency and reduces the risk of convergence in local minima. This simplicity makes Jaya FFNN more robust and practical for real-world applications, particularly in capturing the nonlinear relationships crucial to quadcopter motor dynamics, such as thrust, torque, and

**Table 5** The comparative analysis of proposed model for linearly varying throttle

| Parameter    | Metrics | SVM     | REG_NN  | C_FIT   | FFNN    | PSO_FFNN | Proposed JAYA_FFNN |
|--------------|---------|---------|---------|---------|---------|----------|--------------------|
| SPEED (rpm)  | $R^2$   | 0.8991  | 0.841   | 0.9659  | 0.9316  | 0.9605   | 0.997              |
|              | MAE     | 278.88  | 485     | 248     | 160     | 101.4    | 75.33              |
|              | RMSE    | 440.94  | 562.731 | 325.9   | 308.591 | 124.822  | 103.561            |
| THRUST (kgf) | $R^2$   | 0.8962  | 0.8726  | 0.9848  | 0.9639  | 0.9795   | 0.9951             |
|              | MAE     | 0.215   | 0.04151 | 0.01854 | 0.15    | 0.023    | 0.002              |
|              | RMSE    | 0.245   | 0.0472  | 0.0213  | 0.0328  | 0.0247   | 0.0121             |
| TORQUE (N-m) | $R^2$   | 0.9649  | 0.8984  | 0.976   | 0.8853  | 0.9743   | 0.9844             |
|              | MAE     | 0.0321  | 0.0347  | 0.01179 | 0.0125  | 0.0512   | 0.0018             |
|              | RMSE    | 0.00423 | 0.0064  | 0.013   | 0.0084  | 0.0351   | 0.00398            |
| CURRENT (A)  | $R^2$   | 0.9258  | 0.8743  | 0.9888  | 0.875   | 0.9802   | 0.9985             |
|              | MAE     | 0.204   | 0.49951 | 0.21128 | 0.424   | 0.163    | 0.015              |
|              | RMSE    | 0.234   | 0.573   | 0.222   | 0.7423  | 0.29514  | 0.0814             |
| P_MECH (W)   | $R^2$   | 0.9997  | 0.8637  | 0.9898  | 0.9301  | 0.9682   | 0.992              |
|              | MAE     | 1.688   | – 4.506 | 1.69256 | 1.374   | 0.167    | 0.707              |
|              | RMSE    | 2.11    | 5.1691  | 2.89    | 4.96    | 3.3449   | 1.6722             |

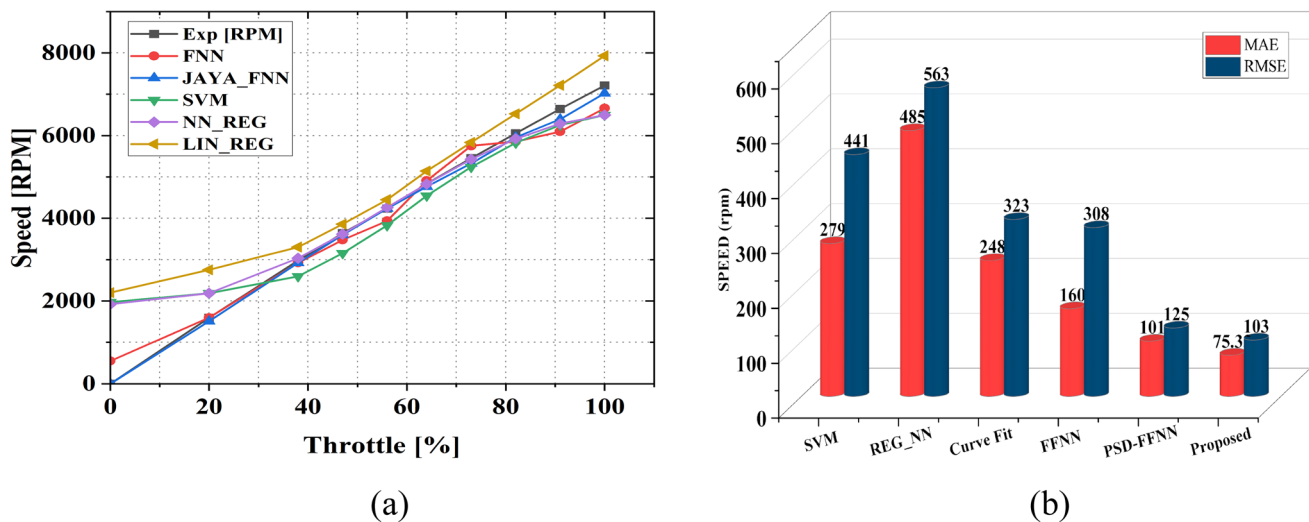


Fig. 12 a. Comparison of different regression models for motor speed. b. MAE and RMSE analysis of predicted speed

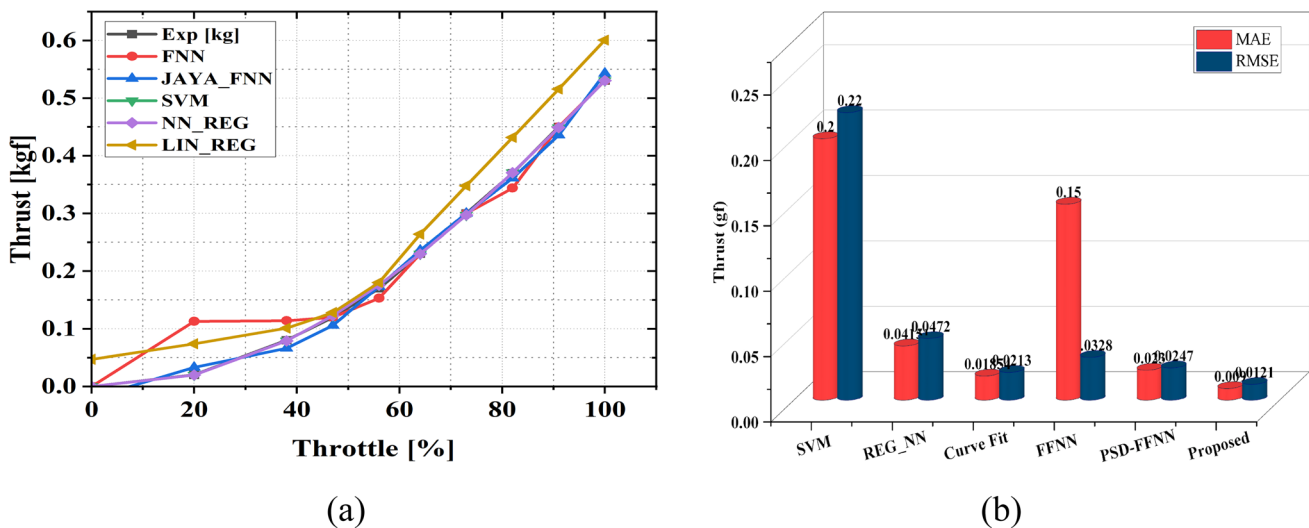


Fig. 13 a. Comparison of different regression models for motor thrust. b. MAE and RMSE analysis of predicted thrust

speed. Table 4 compares the key attributes of Jaya algorithm with PSO algorithm.

With improved predictive accuracy, Jaya FFNN is well-suited for real-time use in quadcopters. It offers stability, ease of deployment, and consistent performance prediction despite rapid changes in motor dynamics, making it a reliable choice for effective motor control and monitoring. The Jaya-optimized FFNN model demonstrates clear advantages in practical applications, combining high accuracy, low parameter dependency, fast convergence, and robustness against local minima. This positions it as a superior model for non-linear, real-time applications, such as motor performance prediction, by providing a stable, accurate, and easily deployable solution compared to other baseline models.

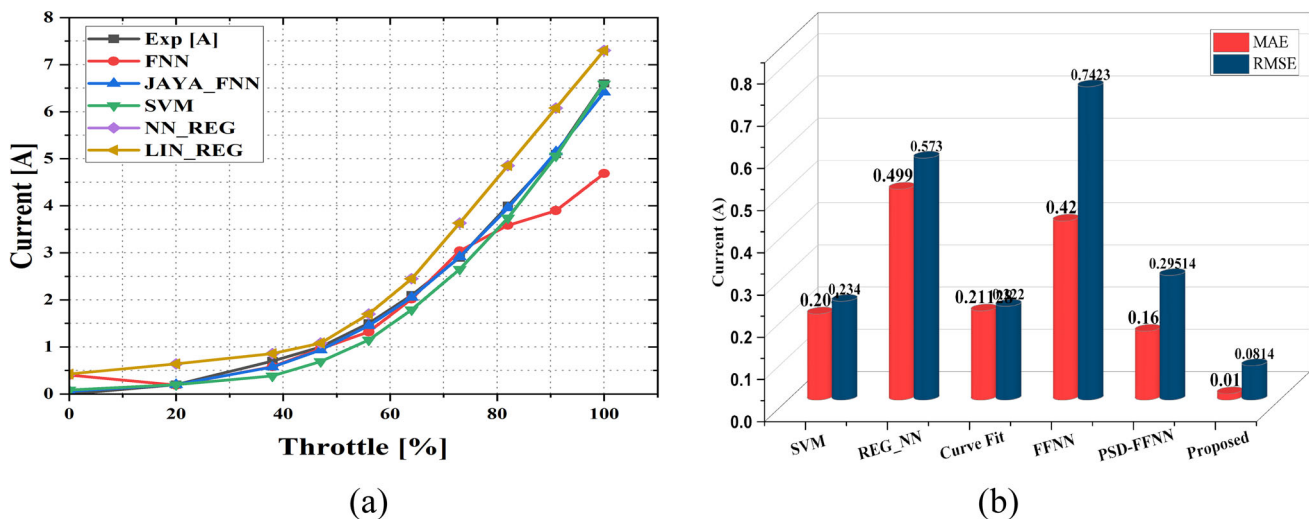
### 3 Experimental validation of the proposed model

The model is developed using MATLAB Simulink and validated with the experimental results. Two sets of data are demonstrated for experimental validation. The first set includes the experimental results of linear step variation of throttle from 0 to 100%. The second set includes the validation of the model using a full flight from takeoff to landing data.

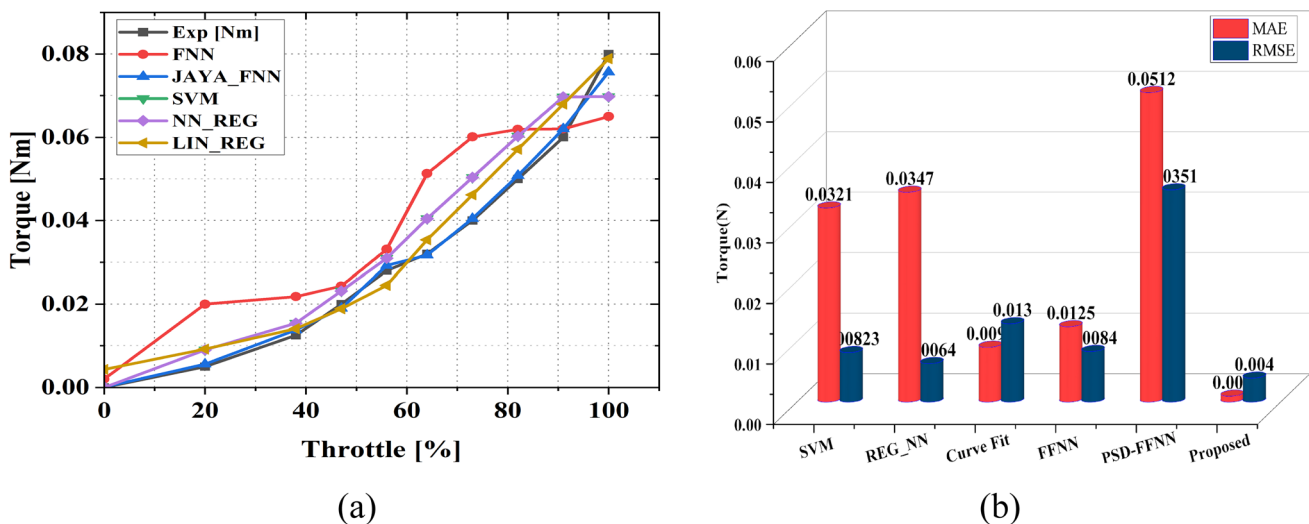
#### 3.1 Linear variation of throttle

With a linear variation of throttle, the model predicts the motor torque (N-m), induced thrust (kgf), current (A), speed





**Fig. 14** **a.** Comparison of different regression models for motor current. **b** MAE and RMSE analysis of predicted current



**Fig. 15** **a** Comparison of different regression models for motor torque. **b.** MAE and RMSE analysis of predicted torque

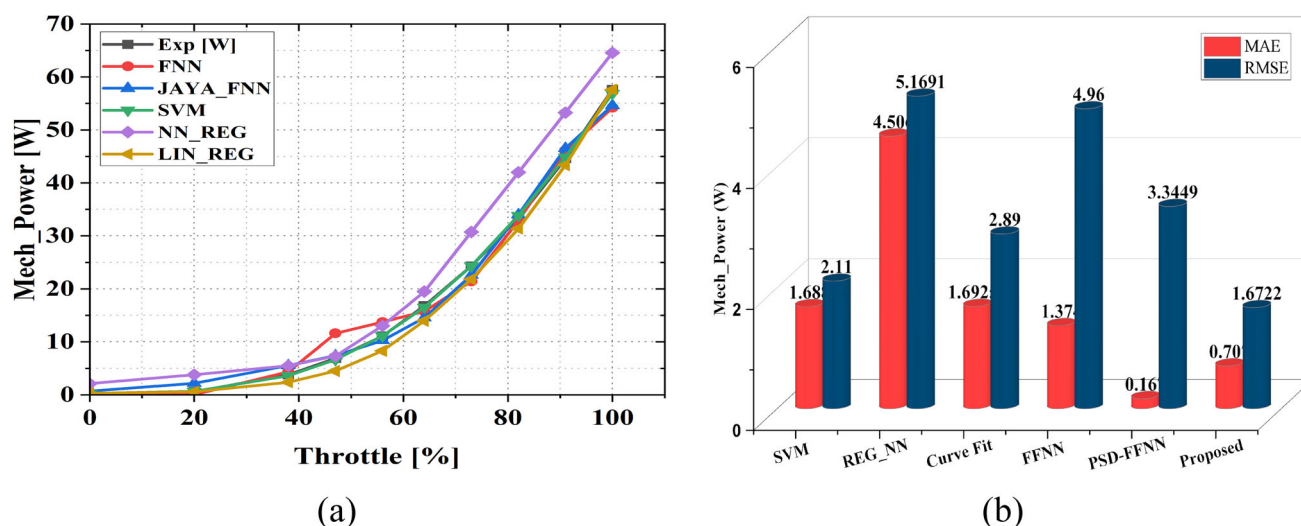
(RPM), and output power (W). The proposed model performance is compared with other models like linear regression (LIN-REG), support vector machine (SVM), neural network regression model (NN-REG), feed forward neural network (FFNN), PSO-FFNN, and verified with experimental data. As per the regression metrics shown in Table 5, the proposed Jaya FFNN model predicts the speed, thrust, torques, current, and mechanical power of quadcopter motor better than the other models with high  $R^2$  values of 0.997, 0.9951, 0.9844, 0.9985, and 0.992, respectively. The corresponding predicted results and the performance parameters are plotted in Figs. 12, 13, 14, 15, and 16.

The comparison of the predicted speed by the regression models with experimental data is shown in Fig. 12a, and the corresponding regression metrics of all the models are shown in Fig. 12b. The proposed model predicts the required

speed for the given throttle input with better accuracy. The MAE and RMSE values of the proposed model are 75.3 rpm and 103 rpm, respectively, whereas the FFNN model has the MAE and RMSE values 160 rpm and 308 rpm.

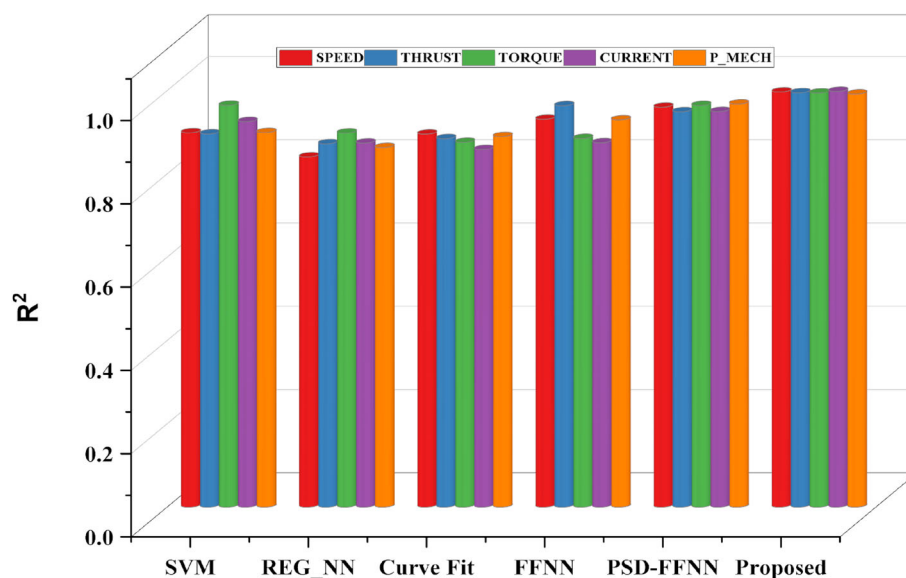
The comparison of the predicted thrust by the regression models with experimental data is shown in Fig. 13a, and the corresponding regression metrics of the ANN models are shown in Fig. 13b. The proposed model predicts the required thrust for the given throttle input with better accuracy. The MAE and RMSE values of the proposed model are 0.002 kgf and 0.0122 kgf, respectively, outperforms the FFNN model with MAE and RMSE values 0.15 kgf and 0.03288 kgf, respectively.

The comparison of the predicted current by the regression models with experimental data is shown in Fig. 14a, and the corresponding regression metrics of the models are shown



**Fig. 16** **a** Comparison of different regression models for motor mechanical power. **b.** MAE and RMSE analysis of predicted mechanical power

**Fig. 17**  $R^2$  analysis of predicted results



**Table 6** Deviation analysis of the optimized model for linearly varying throttle

| S. No. | Parameters | Accuracy (%)  |           |               |           | Overfitting |           |
|--------|------------|---------------|-----------|---------------|-----------|-------------|-----------|
|        |            | Min. throttle |           | Max. throttle |           |             |           |
|        |            | FFNN          | Jaya FFNN | FFNN          | Jaya FFNN | FFNN        | Jaya FFNN |
| 1      | Speed      | 96.2          | 99.6      | 92.3          | 96.4      | Yes         | No        |
| 2      | Thrust     | 35.7          | 98.4      | 89.3          | 97.5      | Yes         | No        |
| 3      | Current    | 94.5          | 97.9      | 71.1          | 97.2      | Yes         | No        |
| 4      | Torque     | 45.5          | 96.9      | 81.2          | 94.9      | Yes         | No        |

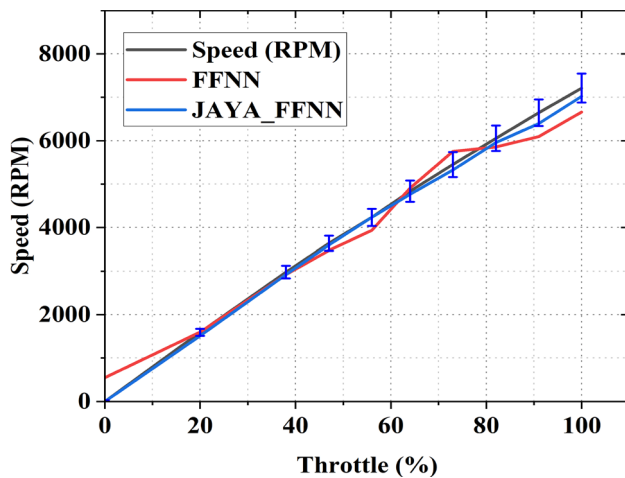


Fig. 18 Deviation analysis of predicted speed

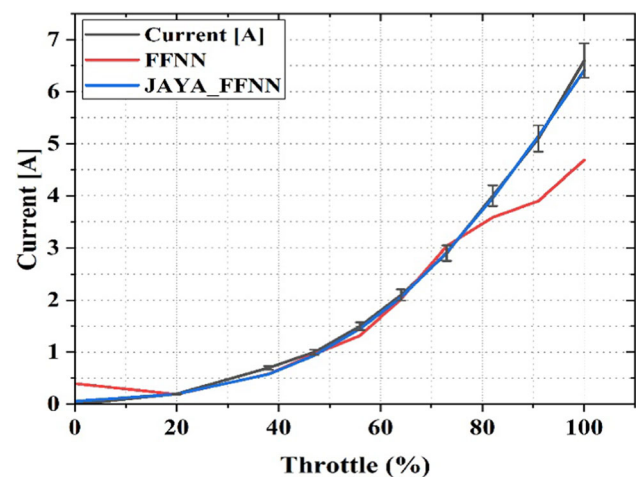


Fig. 20 Deviation analysis of predicted current

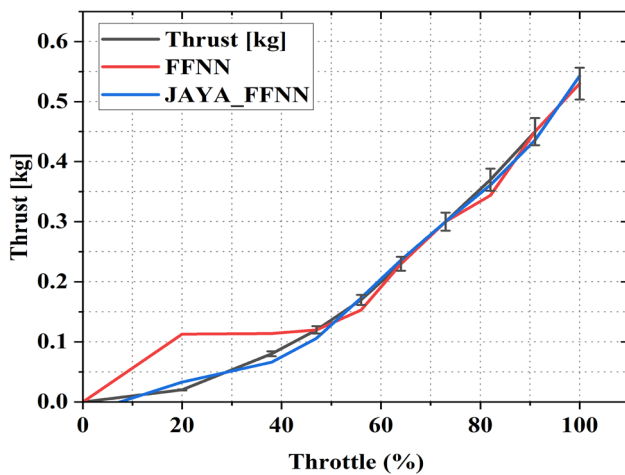


Fig. 19 Deviation analysis of predicted thrust

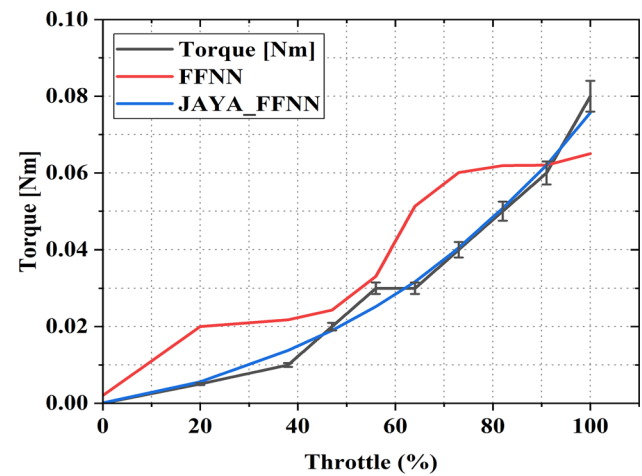


Fig. 21 Deviation analysis of predicted torque

in Fig. 14b. The proposed model excels the existing models predicting the current for the given throttle input. The MAE and RMSE values of the proposed model are 0.015 A and 0.0814 A, respectively, whereas the FFNN model has the MAE and RMSE values 0.424 A and 0.74230 A.

The comparison of the predicted torque by the regression models with experimental data is shown in Fig. 15a, and the corresponding regression metrics of the ANN models are shown in Fig. 15b. The proposed model predicts the required torque for the given throttle input with better accuracy. The MAE and RMSE values of the proposed model are 0.0018 N-m and 0.00398 N-m, respectively, whereas the FFNN model has the MAE and RMSE values 0.0125 N-m 0.0084 N-m.

The comparison of the predicted mechanical power by the regression models with experimental data is shown in Fig. 16a, and the corresponding regression metrics of the ANN models are shown in Fig. 16b. The proposed model predicts the required power for the given throttle input with

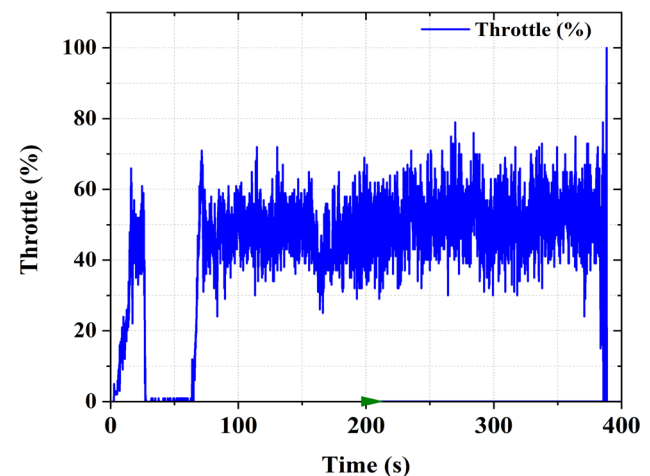


Fig. 22 Throttle input to the experimental setup model

**Table 7** Comparative analysis of a proposed model for full flight from takeoff to landing cycle

| S.No | Parameters   | Regression metrics |       |        |        |        |        |
|------|--------------|--------------------|-------|--------|--------|--------|--------|
|      |              | JAYA–FFNN          |       |        | FFNN   |        |        |
|      |              | MAE                | RMSE  | $R^2$  | MAE    | RMSE   | $R^2$  |
| 1    | Speed (rpm)  | 71.3               | 147.4 | 0.9704 | 197    | 277.16 | 0.9209 |
| 2    | Thrust (kgf) | 3.54               | 8.46  | 0.9737 | 6.05   | 14.04  | 0.8952 |
| 3    | Current (A)  | 0.08               | 0.17  | 0.9715 | 0.25   | 0.34   | 0.8874 |
| 4    | Torque(N-m)  | 0.00075            | 0.067 | 0.9700 | 0.0014 | 0.123  | 0.9099 |

better accuracy. The MAE and RMSE values of the proposed model are 0.707 W and 1.6722 W, respectively, whereas the FFNN model has the MAE and RMSE values 1.374 W and 4.96 W.

The comparison of regression metrics ( $R^2$ ) of the predicted models is compared in Fig. 17. The proposed model predicts with accuracy 0.997, 0.9951, 0.9844, 0.9985, and 0.992 of in predicting motor test results. Whereas the FFNN model has the regression metrics as 0.9316, 0.9639, 0.8853, 0.875, and 0.9301 for predicting speed, thrust, torque, current, and mechanical power, respectively.

### 3.2 Deviation analysis

A deviation analysis is conducted to calculate the accuracy of the model with and without optimization from the experimental results. From the predicted speed versus % throttle plotted in Fig. 18, it is clear that the Jaya optimization gives an accuracy in speed of 99.6% at minimum throttle and 96.4% at maximum throttle without overfitting. The Jaya optimization algorithm enables the FFNN model to achieve higher accuracy in predicting motor speed by efficiently tuning the model's parameters. Its ability to optimize without additional control parameters, avoid local minima, and adapt to different cost functions made it highly effective for this nonlinear problem, leading to precise predictions of speed based on throttle input. The accuracy in the prediction is tabulated in Table 6.

In Fig. 19, it is clear that the Jaya optimization gives an accuracy in thrust of 98.4% at minimum throttle and 97.5% at maximum throttle without overfitting. In Fig. 20, it is clear that the Jaya optimization gives an accuracy in current of 97.9% at minimum throttle and 97.2% at maximum throttle without overfitting. In Fig. 21, it is clear that the Jaya optimization gives an accuracy in torque of 96.9% at minimum throttle and 94.9% at maximum throttle without overfitting. The predicted results show a deviation less than  $\pm 5\%$  from the experimental data.

### 3.3 Validation using full flight from takeoff to landing data

The real-time flight data for full flight from takeoff to landing are collected in our campus from a quadcopter and are fed as the throttle input to the thrust stand as well as the proposed model, which is shown in Fig. 22 and the predicting accuracy is compared in Table 7.

Various performance parameters predicted by the model coincides with the experimental data. Figure 23a shows the comparison of predicted speed from the proposed model with and without Jaya optimization to the experimental result. The error predicting the speed reduces by 12.5% with the inclusion of Jaya optimization as shown in Fig. 23b.

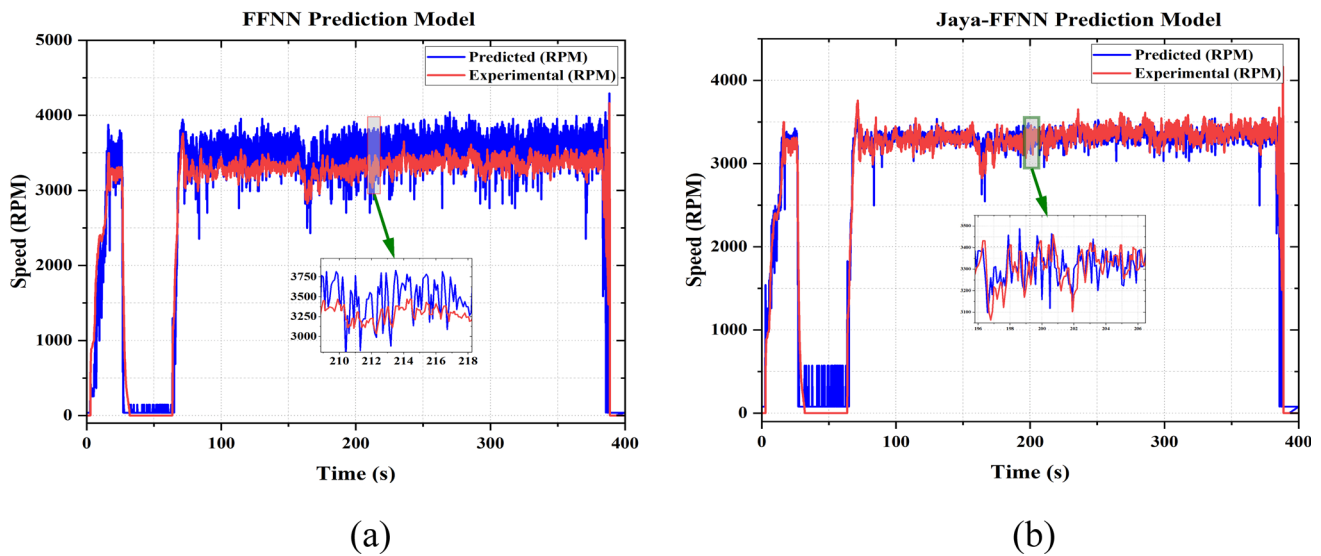
Figure 24a and b shows the comparison of predicted thrust from the proposed model with and without Jaya optimization, respectively, to the experimental result. The error in predicting the thrust reduces by 7.85% with the inclusion of Jaya optimization as shown in Fig. 24b.

Figure 25a and b shows the comparison of predicted current from the proposed model with and without Jaya optimization respectively to the experimental result. The error in predicting the current reduces by 17.5% with the inclusion of Jaya optimization as shown in Fig. 25b.

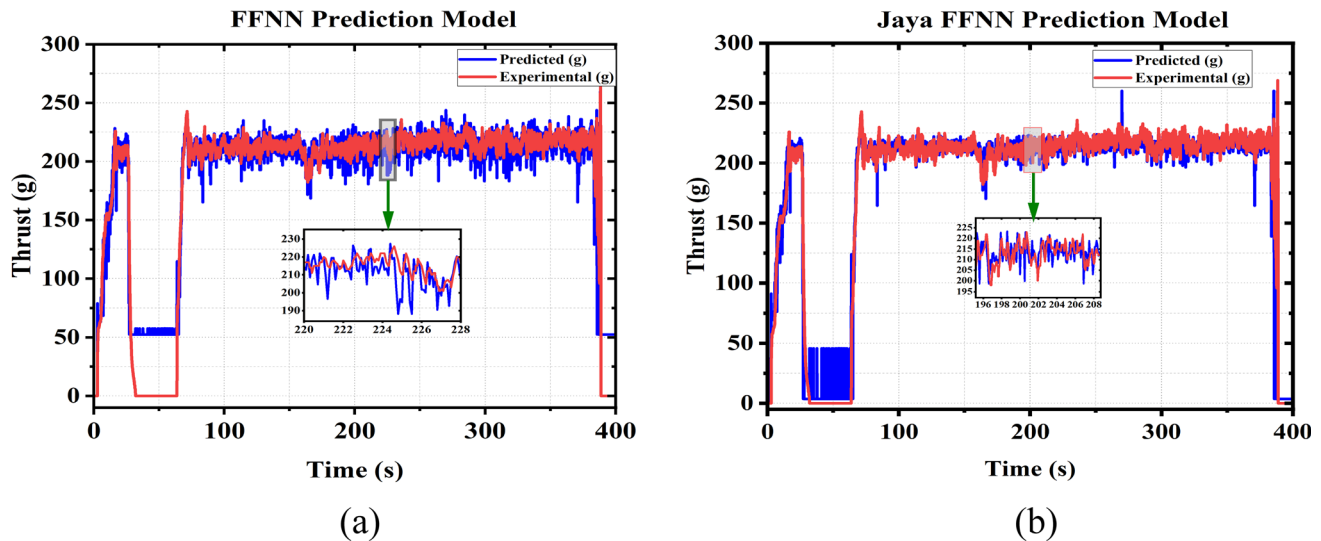
Figure 26a and b shows the comparison of predicted torque from the proposed model with and without Jaya optimization respectively to the experimental result. The error in predicting the torque reduces by 6.49% with the inclusion of Jaya optimization as shown in Fig. 26b.

## 4 Conclusions

In this paper, a data-driven approach to model the performance of a brushless DC (BLDC) motor for quadcopters is proposed using throttle position as the sole input parameter. The throttle position serves as a direct control input, influencing motor RPM and, consequently, thrust, current, torque, and the output power. The analysis demonstrates a strong correlation between throttle position and key performance



**Fig. 23** **a** Experimental validation of predicted speed by FFNN regression model. **b** Experimental validation of predicted speed by Jaya-FFNN regression model

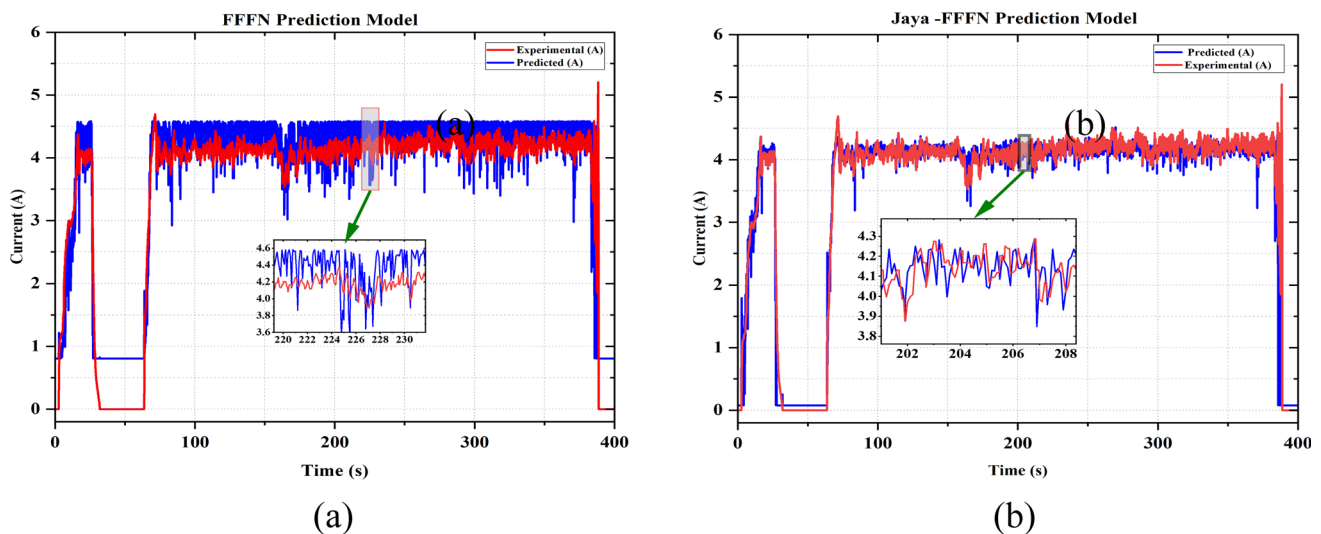


**Fig. 24** **a**. Experimental validation of predicted thrust by FFNN regression model. **b** Experimental validation of predicted speed by Jaya-FFNN regression model

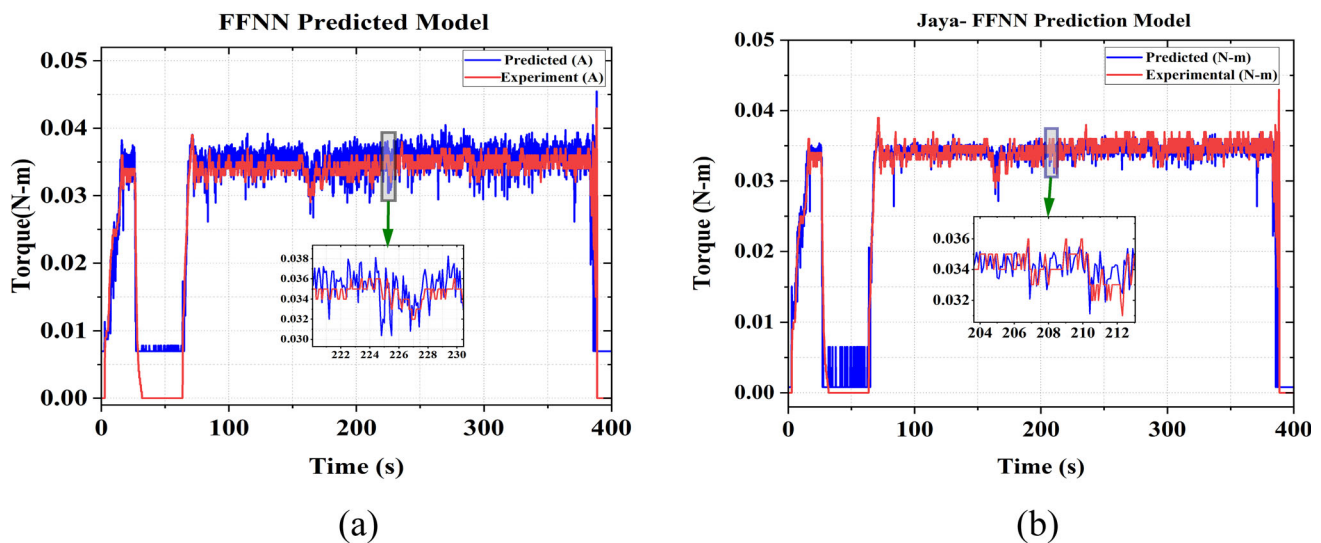
metrics, substantiated by MSE, MAE, RMSE,  $R^2$ , and experimental validation. By employing a Jaya-optimized feed forward neural network (FFNN), high predictive accuracy of  $\pm 5\%$  is obtained for minimum and maximum throttle positions with both linear variation and real-time flight

conditions, showcasing the model's efficacy. This approach simplifies the modeling process, enabling real-time performance predictions with reduced computational complexity. The proposed model can be further used to investigate the hybrid energy management in delivery quadcopters and e-aircraft applications.





**Fig. 25** **a.** Experimental validation of predicted current by FFNN regression model. **b** Experimental validation of predicted current by Jaya-FFNN regression model



**Fig. 26** **a.** Experimental validation of predicted torque FFNN regression model. **b.** Experimental validation of predicted torque Jaya-FFNN regression model

**Acknowledgements** The author would like to thank The Management, Coimbatore Institute of Technology, Coimbatore, Tamilnadu, India – 641014 for providing continuous support to conduct our study and complete the work.

**Author contribution** Conceptualization, methodology, software, visualization were contributed by R.C; analysis, supervision, and discussion of methods and results were involved in R.C, K.S, and B.R.; Writing—original draft preparation and writing—review and editing were performed by R.C. and K.S; All authors have read and agreed to the published version of the manuscript R.C: Chandrasekaran Ravichandran, B.R:Dr.B.Raja, K.S:Dr.K.Selavajothi.

## Declarations

**Conflict of interest** The authors declare no competing interests.

**Data availability** Data will be made available on request from the authors.

## References

1. Idrissi M, Salami M, Annaz F (2022) A review of quadrotor unmanned aerial vehicles: applications architectural design and control algorithms. *J Intell Robot Syst Theory Appl*. <https://doi.org/10.1007/s10846-021-01527-7>
2. Saeed AS, Younes AB, Islam S, Dias J, Seneviratne L, Cai G (2015) A review on the platform design, dynamic modeling and control

- of hybrid UAVs. 2015 Int Conf Unmanned Aircr Syst ICUAS 2015:806–815. <https://doi.org/10.1109/ICUAS.2015.7152365>
3. Nawaz H, Ali HM, Massan S-R (2019) Applications of unmanned aerial vehicles: a review. *C Tecnol Innovación Apl a la pyme* 2019:85–105. <https://doi.org/10.17993/3ctecno.2019.specia.lissu3.85-105>
  4. Carrio A, Sampedro C, Rodriguez-Ramos A, Campoy P (2017) A review of deep learning methods and applications for unmanned aerial vehicles. *J Sens* 2017:3296874. <https://doi.org/10.1155/2017/3296874>
  5. Abro KA, Atangana A, Gómez-Aguilar JF (2023) Chaos control and characterization of brushless DC motor via integral and differential fractal-fractional techniques. *Int J Model Simul* 43(4):416–425. <https://doi.org/10.1080/02286203.2022.2086743>
  6. Esmail MS, Merzban MH, Khalaf AAM, Hamed HFA, Hussein AI (2022) Attitude and altitude nonlinear control regulation of a quadcopter using quaternion representation. *IEEE Access* 10:5884–5894. <https://doi.org/10.1109/ACCESS.2022.3141544>
  7. Kumar R, Agarwal SR, Kumar M (2021) Modeling and control of a tethered tilt-rotor quadcopter with atmospheric wind model. *IFAC-PapersOnLine* 54(20):463–468. <https://doi.org/10.1016/j.ifacol.2021.11.216>
  8. Ye J, Wang J, Song T, Wu Z, Tang P (2021) Nonlinear modeling the quadcopter considering the aerodynamic interaction. *IEEE Access* 9:134716–134732. <https://doi.org/10.1109/ACCESS.2021.3116676>
  9. Farias Filho JG, Dórea CE, Bessa WM, Farias JL (2018) Modeling, test benches and identification of a quadcopter. In: *Proc. - 13th Lat. Am. Robot. Symp. 4th Brazilian Symp. Robot. LARS/SBR 2016*, pp 49–54. <https://doi.org/10.1109/LARS-SBR.2016.67>
  10. Ji R, Ma J, Sam Ge S (2020) Modeling and control of a tilting quadcopter. *IEEE Trans Aerosp Electron Syst* 56(4):2823–2834. <https://doi.org/10.1109/TAES.2019.2955525>
  11. Hentati AI, Fourati LC, Elgharbi E, Tayeb S (2023) Simulation tools, environments and frameworks for UAVs and multi-UAV-based systems performance analysis (version 2.0). *Int J Modelling Simul* 43(4):474–490. <https://doi.org/10.1080/02286203.2022.2092257>
  12. Alabsi MI, Fields TD (2019) Real-time closed-loop system identification of a quadcopter. *J Aircr* 56(1):324–335. <https://doi.org/10.2514/1.C034219>
  13. Lei W, Li C (2017) On-line aerodynamic identification of quadrotor and its application to tracking control. *IET Control Theory Appl* 11(17):3097–3106. <https://doi.org/10.1049/iet-cta.2017.0664>
  14. Sun S, De Visser CC, Chu Q (2019) Quadrotor gray-box model identification from high-speed flight data. *J Aircr* 56(2):645–661. <https://doi.org/10.2514/1.C035135>
  15. Elokda E, Coulson J, Beuchat PN, Lygeros J, Dörfler F (2021) Data-enabled predictive control for quadcopters. *Int J Robust Nonlinear Control* 31(18):8916–8936. <https://doi.org/10.1002/rnc.5686>
  16. Jiang F, Pourpanah F, Hao Q (2020) Design, implementation, and evaluation of a neural-network-based quadcopter UAV system. *IEEE Trans Ind Electron* 67(3):2076–2085. <https://doi.org/10.1109/TIE.2019.2905808>
  17. Al-Mahasneh AJ, Anavatti SG, Garratt MA (2018) Review of applications of generalized regression neural networks in identification and control of dynamic systems, [Online]. Available: <http://arxiv.org/abs/1805.11236>
  18. Ram M, Kazancoglu Y (2021) Special issue on ‘Recent advances in multi-disciplinary modeling & simulations.’ *Int J Modelling Simul* 41(5):325. <https://doi.org/10.1080/02286203.2021.1983080>
  19. Chu Z, Fei Hu, Bentley ES, Kumar S (2022) Model and simulations of multipath bridge routing for inter-swarm UAV communications in EMANE/CORE. *Int J Model Simul* 42(3):485–505. <https://doi.org/10.1080/02286203.2021.1931789>
  20. Li Y, Qin Y, Xu W, Zhang F (2020) Modeling, identification, and control of non-minimum phase dynamics of Bi-copter UAVs. In: *IEEE/ASME Int. Conf. Adv. Intell. Mechatronics, AIM*, vol. 2020-July, pp 1249–1255. <https://doi.org/10.1109/AIM43001.2020.9158910>
  21. Elragal HM (2009) Improving neural networks prediction accuracy using particle swarm optimization combiner. *Int J Neural Syst* 19(5):387–393. <https://doi.org/10.1142/S0129065709002099>
  22. Mosavi MR, Khishe M (2017) Training a feed-forward neural network using particle swarm optimizer with autonomous groups for sonar target classification. *J Circuits Syst Comput* 26(11):1–20. <https://doi.org/10.1142/S0218126617501857>
  23. Ansari MS, Jalil MF, Bansal RC (2023) A review of optimization techniques for hybrid renewable energy systems. *Int J Modelling Simul* 43(5):722–735. <https://doi.org/10.1080/02286203.2022.2119524>
  24. Hernández JD, Nandar JE, Changoluisa ID, Cruz PJ, Valencia E (2021) Test-bench development for the efficiency analysis of UAV motor-propeller sets. In: *ETCM 2021 - 5th Ecuador Tech. Chapters Meet*, pp 4–9. <https://doi.org/10.1109/ETCM53643.2021.9590717>
  25. Touil DE, Terki N, Hamiane M, Aouina A, Brahim KS (2022) Image-based visual servoing control of a quadcopter air vehicle. *Int J Model Simul* 42(2):203–216. <https://doi.org/10.1080/02286203.2021.1871719>
  26. Yu Y, Ding X (2012) A quadrotor test bench for six degree of freedom flight. *J Intell Robot Syst Theory Appl* 68(3–4):323–338. <https://doi.org/10.1007/s10846-012-9680-y>
  27. Mathew R, Hiremath SS (2021) Control of stepper motor driven differential drive robot for tracking waypoint trajectories: modeling, simulation, optimization and experimental investigations. *Int J Model Simul* 41(3):176–194. <https://doi.org/10.1080/02286203.2019.1700347>
  28. Bohat VK, Arya KV (2018) An effective gbest-guided gravitational search algorithm for real-parameter optimization and its application in training of feedforward neural networks. *Knowl-Based Syst* 143:192–207. <https://doi.org/10.1016/j.knosys.2017.12.017>
  29. Maulud D, Abdulazeez AM (2020) A review on linear regression comprehensive in machine learning. *JASTT* 1(2):140–147. <https://doi.org/10.38094/jastt1457>
  30. Alkorbi AS, Tanveer M, Shahid H, Qadir MB, Ahmad F, Khaliq Z, Jalalah M, Irfan M, Algadi H, Harraz FA (2024) Comparative analysis of feed-forward neural network and second-order polynomial regression in textile wastewater treatment efficiency. *AIMS Math* 9(5):10955–10976. <https://doi.org/10.3934/math.2024536>
  31. Roy A, Chakraborty S (2023) Support vector machine in structural reliability analysis: a review. *Reliab Eng Syst Saf* 233:109126. <https://doi.org/10.1016/j.ress.2023.109126>
  32. Sazli M (2006) A brief review of feed-forward neural networks. *Commun Fac Sci Univ Ank* 50:11–17. [https://doi.org/10.1501/commua1-2\\_00000000026](https://doi.org/10.1501/commua1-2_00000000026)
  33. Sundarajoo S, Soomro DM (2023) A particle swarm optimization trained feedforward neural network for under-voltage load shedding. *ECTI Trans Electr Eng Electron Commun* 21(2):1–16. <https://doi.org/10.37936/ecti-ee.2023212.249825>
  34. Debata PP, Mohapatra P (2021) A hybrid convolutional neural network approach for feature selection and disease classification. *Turk J Electr Eng Comput Sci* 29(8):2580–2599. <https://doi.org/10.3906/elk-2105-43>
  35. Garro BA, Vázquez RA (2015) Designing artificial neural networks using particle swarm optimization algorithms. *Comput Intell Neurosci*. <https://doi.org/10.1155/2015/369298>
  36. Venkata Rao R (2016) Jaya: A simple and new optimization algorithm for solving constrained and unconstrained optimization problems. *Int J Ind Eng Comput* 7(1):19–34. <https://doi.org/10.5267/j.ijiec.2015.8.004>

37. Majji R, Nalinipriya G, Vidyadhari C, Cristin R (2021) Jaya ant lion optimization-driven deep recurrent neural network for cancer classification using gene expression data. *Med Biol Eng Comput* 59(5):1005–1021. <https://doi.org/10.1007/s11517-021-02350-w>
38. Uzlu E (2019) Application of Jaya algorithm-trained artificial neural networks for prediction of energy use in the nation of Turkey. *Energy Sour Part B Econ Plan Policy* 14(5):183–200. <https://doi.org/10.1080/15567249.2019.1653405>

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.